

Mahi Aminu Aliyu

Abuja, Nigeria

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RESEARCH INTERESTS

My research focuses on how machine learning and deep learning models learn meaningful representations, with a particular emphasis on causal representation learning and inductive biases.

EDUCATION

M.Sc., Computer Science, Abubakar Tafawa Balewa University, Nigeria 2023 – Present

Advisors: Dr Badamsi Imam Ya'u 📧

B.Sc., Computer Science, Al-Qalam University Katsina, Nigeria 2018 – 2023

Thesis: Water Quality Prediction Model

Advisor: Dr Mansir Abubakar 📧

EXPERIENCE

Artificial Intelligence and Social Informatics Associate 2025 – Present

Africa Hub for Innovation and Development

- Refactored WellBridge (a Tuberculosis chatbot) into class-based architecture with GPU-accelerated STT, TTS, and LLM components, improving maintainability, scalability, and performance (30%).
- Expanded its reach through WhatsApp integration and established a structured feedback loop with AHFID staff to guide system improvements.
- Scraped and analyzed 2,000+ public health-related social media posts across four platforms to build an analysis-ready dataset.

Independent Researcher 2024 – Present

ML Collective (Remote)

- Led the ML Theory and Optimization group, conducting literature-driven research on generalization techniques with a focus on causal representation learning, including experimental design, execution, and research reporting.
- Contributed to multimodal research by curating, cleaning, and organizing datasets to support reliable experimentation and analysis.
- Presented research ideas and findings at conferences and workshops, engaging with the research community and integrating feedback into ongoing work.
- Anchored weekly Saturday reading sessions, guiding technical paper discussions and improving critical paper digestion across the community.

Data Engineer & Analyst 2023 – 2025

Horizon IT Services

- Curated a low-resource Hausa dataset by extracting and cleaning 500+ text entries from Al-Barka radio station's news archives
- Applied advanced data cleaning techniques to transform 2,000+ raw Hausa news entries into a structured CSV dataset
- Collaborated on dataset enhancement by introducing categorization and optional labeling, improving data usability by 30%

PUBLICATIONS & MANUSCRIPTS

[1] **Mahi Aliyu**, Mansir Abubakar, Armaya'u Z. Umar, Alwatben Batoul Rashed, and Rukayya Musa Ismail. Analysis of machine learning models for efficient water quality prediction. In Journal of Information and Technology Management (JISTM), 10(40), Sep. 2025. 📧

[2] Mardiyah Oduwale, Prince Mireku, Fatimo Adebajo, Oluwatosin Olajide, **Mahi Aminu Aliyu**, Jekaterina Novikova. AfriCaption: Establishing a New Paradigm for Image Captioning in African Languages. arXiv preprint 📧

[3] **Mahi Aminu Aliyu**, Chisom Chibuike, Fatimo Adebajo, Omokolade Awosanya, Samuel Oyeneye. Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation. arXiv preprint 📧

CONFERENCES & SUMMER SCHOOLS

Deep Learning Indaba 2025, Rwanda – Attendee Selected to attend African largest AI conference with travel grant acquired from ML Collective.	2025
Machine Learning Summer School - Senegal (MLSS) – Participant Selected to attend advanced ML training led by top global researchers	2025
Deep Learning IndabaX, Nigeria - Attendee Participated in African-led ML training, discussions, and community-building events	2024

SELECTED PROJECTS

Cross-Lingual Aspect-Based Sentiment Analysis in African Languages - Translated SemEval-2014 to 17 African languages (afri-SemEval-2014) with aspect alignment heuristics. - Modified and fine-tuned pretrained transformer architectures (e.g., XLM-R, AfroXLM-R) for causal reasoning for aspect-based sentiment analysis tasks. - Causal models showed superior out-of-distribution (OOD) performance with fewer training steps.	
SageBot - Built AI tool that summarizes technical docs for developers, cutting review time by 15%. - Leveraged LLMs (ChatGPT, LLaMA-3), Hugging Face, and Langchain for high summarization accuracy. - Improved developer workflow efficiency by enabling 25% faster comprehension of complex documents.	
Dono Health - Led development of inclusive insurance MVP with Health Center, Vendor, and Insured modules. - Designed serverless AWS backend with Lambda, API Gateway, and CDK. - Integrated Flutterwave for payments and Africa’s Talking USSD API for voucher-based funding.	

TALKS & PRESENTATIONS

Deep Learning Inbada ’25 Poster presentation: "Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation"	2025
MLSS-Senegal ’25 Poster presentation: "Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation"	2025
ML Collective #25 Research Jam Oral Presentation: "Pretrained Models for Water-based Applications"	2025

COMMUNITY SERVICE

City Lead , Data Science Nigeria - Organize AI workshops, hands-on sessions, and reading groups for 150+ community members - Promote AI adoption and capacity building across underserved sectors	2024 – Present
DeepLearning.ai Ambassador - Facilitated knowledge sharing and engaged 100+ peers as an active community ambassador - Hosted "Growing an AI Community" event in Bauchi with 50+ participants - Organized and participated in 5 cross-community events to strengthen the regional AI ecosystem	2023 – Present
Lead , Google Developer Student Clubs - Led the Google Developer Student Club (GDSC) at Al-Qalam University, Katsina, where I with core-team members, organized and facilitated10+ technical workshops and hands-on sessions, reaching and training over 350 students in areas such as data science, web development, and android development. - Applied Google technologies to address UN SDGs, delivering solutions to 5 local communities	2021 – 2022

TECHNICAL SKILLS

Programming Languages: Python, C++, R
ML/AI Frameworks: PyTorch, TensorFlow, Hugging Face, Scikit-learn
Web Frameworks & Tools: Django, Flask, Git/GitHub
Cloud Platforms: Microsoft Azure, AWS & GCP
Operating Systems: Linux
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

RELEVANT COURSES

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|---|------|
| - Machine Learning Specialization, Coursera – DeepLearning.ai & Stanford | 2024 |
| - Statistical Inference, Coursera – Johns Hopkins University | 2023 |
| - Data Science: Foundations using R Specialization, Coursera – Johns Hopkins University | 2022 |

REFERENCES

Dr. Badamasi Imam Ya'u

Senior Lecturer, Abubakar Tafawa Balewa University

Email: biyau@atbu.edu.ng

Dr. Mansir Abubakar

Senior Lecturer, Universiti Teknologi MARA (UiTM),

Email: mansir@uitm.edu.my

Riccardo Cadei

PhD Candidate, Institute of Science and Technology Austria (ISTA)

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